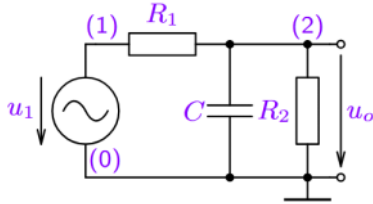


AC5



$$\frac{1}{s+a} \xrightarrow{\mathcal{L}^{-1}} e^{-at} \mathbb{1}(t)$$

$$\frac{1}{s} \longrightarrow \mathbb{1}(t)$$

$$w(t \rightarrow 0) = \lim_{s \rightarrow \infty} H(s) = 0$$

$$w(t \rightarrow \infty) = \lim_{s \rightarrow 0} H(s) = \frac{R_2}{R_1 + R_2}$$

$$h(t \rightarrow 0) = \lim_{s \rightarrow \infty} s H(s) = \lim_{s \rightarrow \infty} \frac{s R_2}{R_1 + R_2 + s R_1 R_2 C} = \frac{1}{R_1 C}$$

$$h(t \rightarrow \infty) = \lim_{s \rightarrow 0} s H(s) = \frac{s R_2}{R_1 + R_2 + s R_1 R_2 C} = 0 \text{ Hz}$$

$$H(s) = \frac{\frac{R_2}{sC}}{R_1 + \frac{R_2}{1+sR_2C}} = \frac{R_2}{R_1 + R_2 + s R_1 R_2 C} = \frac{R_2}{R_1 + R_2 + s R_1 R_2 C}$$

$$= \frac{1}{R_1 C} \cdot \frac{1}{\frac{R_1 + R_2}{R_1 R_2 C} + s} \xrightarrow{\mathcal{L}^{-1}} h(t) = \frac{1}{R_1 C} e^{-\frac{R_1 + R_2}{R_1 R_2 C} t} \mathbb{1}(t)$$

$$W(s) = \frac{H(s)}{s} = \frac{1}{s} \cdot \frac{1/R_1 C}{s + \frac{R_1 + R_2}{R_1 R_2 C}} = \frac{1}{s} \left(\frac{A}{s} + \frac{B}{s + \frac{R_1 + R_2}{R_1 R_2 C}} \right) = \left(\frac{\frac{R_1 R_2 C}{R_1 + R_2}}{s} - \frac{\frac{R_1 R_2 C}{R_1 + R_2}}{s + \frac{R_1 + R_2}{R_1 R_2 C}} \right)$$

$$\mathcal{L}^{-1} \rightarrow w(t) = \frac{R_2}{R_1 + R_2} \mathbb{1}(t) - \frac{R_2}{R_1 + R_2} e^{-\frac{R_1 + R_2}{R_1 R_2 C} t} \mathbb{1}(t)$$

$$= \frac{R_2}{R_1 + R_2} \left(1 - e^{-\frac{R_1 + R_2}{R_1 R_2 C} t} \right) \mathbb{1}(t)$$

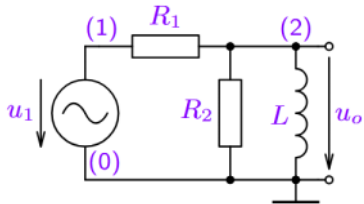
$$1 = A \left(s + \frac{R_1 + R_2}{R_1 R_2 C} \right) + B s$$

$$A + B = 0$$

$$A = \frac{R_1 R_2 C}{R_1 + R_2}$$

$$B = -\frac{R_1 R_2 C}{R_1 + R_2}$$

AC6



$$h(t \rightarrow 0) = \lim_{s \rightarrow \infty} s H(s) = \infty$$

$$h(t \rightarrow \infty) = \lim_{s \rightarrow 0} s H(s) = 0$$

$$w(t \rightarrow 0) = \lim_{s \rightarrow \infty} H(s) = \frac{R_2}{R_1 + R_2}$$

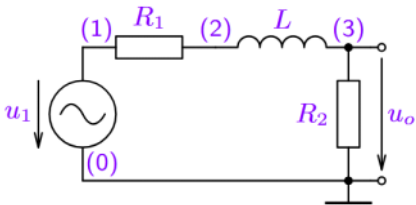
$$w(t \rightarrow \infty) = \lim_{s \rightarrow 0} H(s) = 0$$

$$H(s) = \frac{\frac{s R_2 L}{R_2 + sL}}{R_1 + \frac{s R_2 L}{R_2 + sL}} = \frac{s R_2 L}{R_1 R_2 + s(R_1 + R_2)L} = \frac{R_2}{R_1 + R_2} - \frac{R_2}{R_1 R_2 + s(R_1 + R_2)L}$$

$$= \frac{R_2}{R_1 + R_2} - \frac{R_2}{(R_1 + R_2)^2 L} \frac{1}{\frac{R_1 R_2}{(R_1 + R_2)^2} + s} \rightarrow h(t) = \frac{R_2}{R_1 + R_2} \delta(t) - \frac{R_2}{(R_1 + R_2)^2 L} e^{-\frac{R_1 R_2}{(R_1 + R_2)^2 L} t} \mathbb{1}(t)$$

$$W(s) = \frac{H(s)}{s} = \frac{R_2 L}{R_1 R_2 + s(R_1 + R_2)L} \rightarrow w(t) = \frac{R_2}{R_1 + R_2} e^{-\frac{R_1 R_2}{(R_1 + R_2)^2 L} t} \mathbb{1}(t)$$

AC7



$$H(s) = \frac{R_2}{R_1 + R_2 + sL} = \frac{R_2}{L} \frac{1}{\frac{R_1 + R_2}{L} + s} \rightarrow h(t) = \frac{R_2}{L} e^{-\frac{R_1 + R_2}{L} t} \mathbb{1}(t)$$

$$W(s) = \frac{R_2}{s(R_1 + R_2 + sL)} = \frac{A}{s} + \frac{B}{s + \frac{R_1 + R_2}{L}} = \frac{R_2}{s} - \frac{\frac{R_2 L}{R_1 + R_2}}{s + \frac{R_1 + R_2}{L}}$$

$$R_2 = A(R_1 + R_2 + sL) + B s$$

$$A = \frac{R_2}{R_1 + R_2}$$

$$B = -\frac{R_2 L}{R_1 + R_2}$$

$$w(t) = \frac{R_2}{R_1 + R_2} \mathbb{1}(t) - \frac{R_2}{R_1 + R_2} e^{-\frac{R_1 + R_2}{L} t} \mathbb{1}(t)$$

$$= \frac{R_2}{R_1 + R_2} \left(1 - e^{-\frac{R_1 + R_2}{L} t} \right) \mathbb{1}(t)$$

$$w(t \rightarrow 0) = \lim_{s \rightarrow \infty} H(s) = \lim_{s \rightarrow \infty} \frac{R_2}{R_1 + R_2 + sL} = 0$$

$$w(t \rightarrow \infty) = \lim_{s \rightarrow 0} H(s) = \lim_{s \rightarrow 0} \frac{R_2}{R_1 + R_2 + sL} = \frac{R_2}{R_1 + R_2}$$

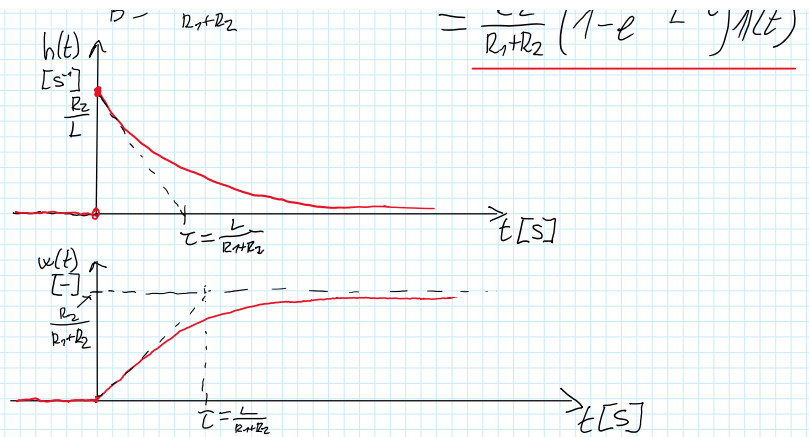
$$h(t) \uparrow$$

$$W(t \rightarrow 0) = \lim_{s \rightarrow \infty} H(s) = \lim_{s \rightarrow \infty} \frac{R_2}{R_1 + R_2 + sL} = 0$$

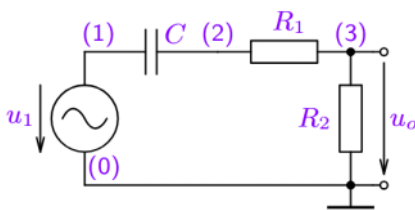
$$W(t \rightarrow \infty) = \lim_{s \rightarrow 0} H(s) = \lim_{s \rightarrow 0} \frac{R_2}{R_1 + R_2 + sL} = \frac{R_2}{R_1 + R_2}$$

$$h(t \rightarrow 0) = \lim_{s \rightarrow \infty} sH(s) = \lim_{s \rightarrow \infty} \frac{sR_2}{R_1 + R_2 + sL} = \frac{R_2}{L}$$

$$h(t \rightarrow \infty) = \lim_{s \rightarrow 0} sH(s) = \lim_{s \rightarrow 0} \frac{sR_2}{R_1 + R_2 + sL} = 0$$



AC8



$$H(s) = \frac{R_2}{R_1 + R_2 + \frac{1}{sC}} = \frac{R_2}{\frac{s(R_1 + R_2)C + 1}{sC}} = \frac{sR_2C}{1 + s(R_1 + R_2)C} = \frac{R_2}{R_1 + R_2} \cdot \frac{\frac{R_2}{R_1 + R_2}}{1 + s(R_1 + R_2)C}$$

$$h(t) = \frac{R_2}{R_1 + R_2} \delta(t) - \frac{R_2}{(R_1 + R_2)^2 C} e^{-\frac{1}{(R_1 + R_2)C} t} \mathbb{1}(t)$$

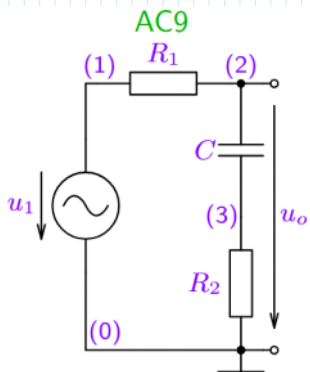
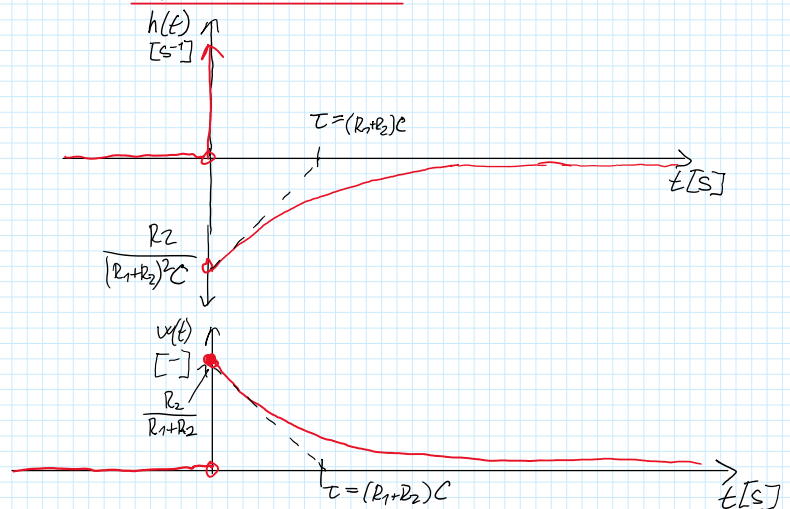
$$W(s) = \frac{R_2 C}{1 + s(R_1 + R_2)C} \quad W(t) = \frac{R_2}{R_1 + R_2} e^{-\frac{1}{(R_1 + R_2)C} t} \mathbb{1}(t)$$

$$h(t \rightarrow 0) = \lim_{s \rightarrow \infty} sH(s) = \lim_{s \rightarrow \infty} \frac{sR_2}{R_1 + R_2 + \frac{1}{sC}} = \infty$$

$$h(t \rightarrow \infty) = \lim_{s \rightarrow 0} sH(s) = \lim_{s \rightarrow 0} \frac{sR_2}{R_1 + R_2 + \frac{1}{sC}} = 0$$

$$w(t \rightarrow 0) = \lim_{s \rightarrow \infty} H(s) = \lim_{s \rightarrow \infty} \frac{R_2}{R_1 + R_2 + \frac{1}{sC}} = \frac{R_2}{R_1 + R_2}$$

$$w(t \rightarrow \infty) = \lim_{s \rightarrow 0} H(s) = \lim_{s \rightarrow 0} \frac{R_2}{R_1 + R_2 + \frac{1}{sC}} = 0$$



$$H(s) = \frac{R_2 + \frac{1}{sC}}{R_1 + R_2 + \frac{1}{sC}} = \frac{1 + sR_2C}{1 + s(R_1 + R_2)C} = \frac{R_2}{R_1 + R_2} + \frac{\frac{R_1}{R_1 + R_2}}{1 + s(R_1 + R_2)C}$$

$$h(t) = \frac{R_2}{R_1 + R_2} \delta(t) + \frac{R_1}{(R_1 + R_2)^2 C} e^{-\frac{1}{(R_1 + R_2)C} t} \mathbb{1}(t)$$

$$W(s) = \frac{1 + sR_2C}{s(1 + s(R_1 + R_2)C)} = \frac{A}{s} + \frac{B}{1 + s(R_1 + R_2)C} = \frac{1}{s} - \frac{R_1 C}{1 + s(R_1 + R_2)C}$$

$$w(t) = \mathbb{1}(t) - \frac{R_1}{(R_1 + R_2)} e^{-\frac{1}{(R_1 + R_2)C} t} \mathbb{1}(t)$$

$$1 + sR_2C = A(1 + s(R_1 + R_2)C) + BS$$

$$A = 1$$

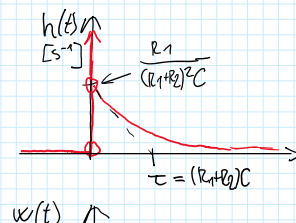
$$B = -R_1 C$$

$$h(t \rightarrow 0) = \lim_{s \rightarrow \infty} sH(s) = \lim_{s \rightarrow \infty} \frac{s(1 + sR_2C)}{1 + s(R_1 + R_2)C} = \infty$$

$$h(t \rightarrow \infty) = \lim_{s \rightarrow 0} sH(s) = \lim_{s \rightarrow 0} \frac{s(1 + sR_2C)}{1 + s(R_1 + R_2)C} = 0$$

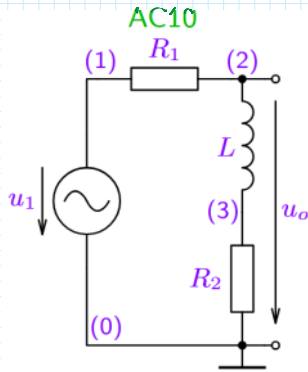
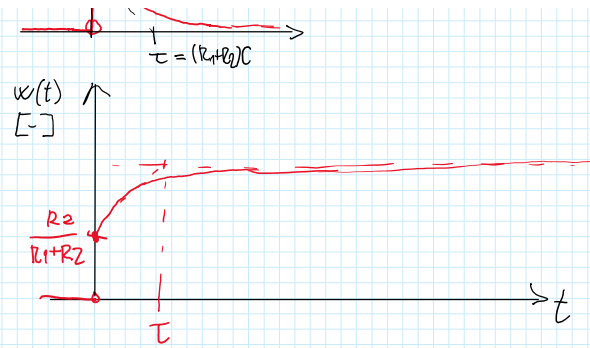
$$w(t \rightarrow 0) = \lim_{s \rightarrow \infty} H(s) = \lim_{s \rightarrow \infty} \frac{1 + sR_2C}{1 + s(R_1 + R_2)C} = \frac{R_2}{R_1 + R_2}$$

$$w(t \rightarrow \infty) = \lim_{s \rightarrow 0} H(s) = \lim_{s \rightarrow 0} \frac{1 + sR_2C}{1 + s(R_1 + R_2)C} = 1$$



$$w(t \rightarrow 0) = \lim_{s \rightarrow \infty} H(s) = \lim_{s \rightarrow \infty} \frac{1}{1+s(R_1+R_2)C} = \frac{1}{R_1+R_2}$$

$$w(t \rightarrow \infty) = \lim_{s \rightarrow 0} H(s) = \lim_{s \rightarrow 0} \frac{1+sR_2C}{1+s(R_1+R_2)C} = 1$$



$$H(s) = \frac{R_2 + sL}{R_1 + R_2 + sL} = 1 - \frac{R_1}{R_1 + R_2 + sL}$$

$$h(t) = \delta(t) - \frac{R_1}{L} e^{-\frac{R_1+R_2}{L}t} \mathbb{1}(t)$$

$$W(s) = \frac{R_2 + sL}{s(R_1 + R_2 + sL)} = \frac{A}{s} + \frac{B}{R_1 + R_2 + sL} = \frac{R_2}{R_1 + R_2} \frac{1}{s} + \frac{R_1}{(R_1 + R_2) \frac{R_1 + R_2 + sL}{L}}$$

$$w(t) = \frac{R_2}{R_1 + R_2} \mathbb{1}(t) + \frac{R_1}{R_1 + R_2} e^{-\frac{R_1+R_2}{L}t} \mathbb{1}(t)$$

$$R_2 + sL = 1(R_1 + R_2 + sL) =$$

$$A = \frac{R_2}{R_1 + R_2}$$

$$L = AL + B$$

$$L - AL = B$$

$$L \left(1 - \frac{R_2}{R_1 + R_2}\right) = B = \frac{R_1 L}{R_1 + R_2}$$

$$h(t \rightarrow 0) = \lim_{s \rightarrow \infty} sH(s) = \lim_{s \rightarrow \infty} \frac{s(R_2 + sL)}{R_1 + R_2 + sL} = \infty$$

$$h(t \rightarrow \infty) = \lim_{s \rightarrow 0} sH(s) = \lim_{s \rightarrow 0} \frac{s(R_2 + sL)}{R_1 + R_2 + sL} = 0$$

$$w(t \rightarrow 0) = \lim_{s \rightarrow \infty} H(s) = \lim_{s \rightarrow \infty} \frac{R_2 + sL}{R_1 + R_2 + sL} = 1$$

$$w(t \rightarrow \infty) = \lim_{s \rightarrow 0} H(s) = \lim_{s \rightarrow 0} \frac{R_2 + sL}{R_1 + R_2 + sL} = \frac{R_2}{R_1 + R_2}$$

